2019

YEAR 10 SCIENCE

James Ruse Agricultural High School

YEARLY EXAMINATION

General Instructions

- Reading time – 5 minutes
- Working time – 95 minutes
- Write using black pen.
- Draw diagrams using pencil.
- NESA approved calculators may be used.
- A periodic table of elements provided.
- Write your name at the top of each separate section booklet.

Total Marks – 80

Section I – Biology

20 marks

Section II – Chemistry

20 marks

Section III – Physics

20 marks

Section IV – Earth and Space

20 marks

Section I – Biology

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions 5 marks

1. What group of organisms are responsible for the breakdown and recycle of matter?

A Herbivores
B Omnivores
C Detritivores
D Carnivores

2. Which two organ systems regulate and maintain homeostasis?

A The nervous and endocrine systems.
B The cardiovascular and integumentary systems.
C The cardiovascular and respiratory systems.
D The respiratory and muscular systems.

3. Two plants of the same species are crossed by a plant breeder. Both plants are pure breeds for flower colour. All the offspring plants have the same flower colour, but originate from different parents.

What type of inheritance does this example demonstrate?

A Sex-linked.
B Recessive
C Dominance
D Co-dominance

4. Rosalind Franklin was an English Chemist and an X-ray crystallographer.

Her work on DNA contributed to the discovery of which feature?

A Long stranded molecule.
B Double helix shape
C Sugar-phosphate backbone
D Complementary nucleotides

5. Over the past several decades, natural selection has caused populations of *Staphylococcus aureus* (an infectious wound bacterium) to evolve resistance to most antibiotics.

If the usage of antibiotics was stopped, what is most likely to happen to the populations of *S. aureus*?

A They will go extinct without the antibiotic.
B The frequency of resistant forms will increase in these populations.
C The populations will begin colonizing new environments.
D The frequency of non-resistant forms will increase in these populations.

Part B – Extended Response Questions 15 marks

Question 6 (2 mark)

The human body has many lines of defence to combat disease.

Identify the first and second lines of defence. **2**

First: ……

Second: ……

Question 7 (3 marks)

A particular species of guinea pigs is known to have either black or white coloured fur, where the black fur (B) is dominant over white fur (b). A homozygous black guinea pig is crossed with a heterozygous black guinea pig.

Construct a punnett square to determine the probability of offspring with the same phenotype as the parents. **3**

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Question 8 (3 marks)

Describe an interspecific interaction between organisms using ONE named example. **3**

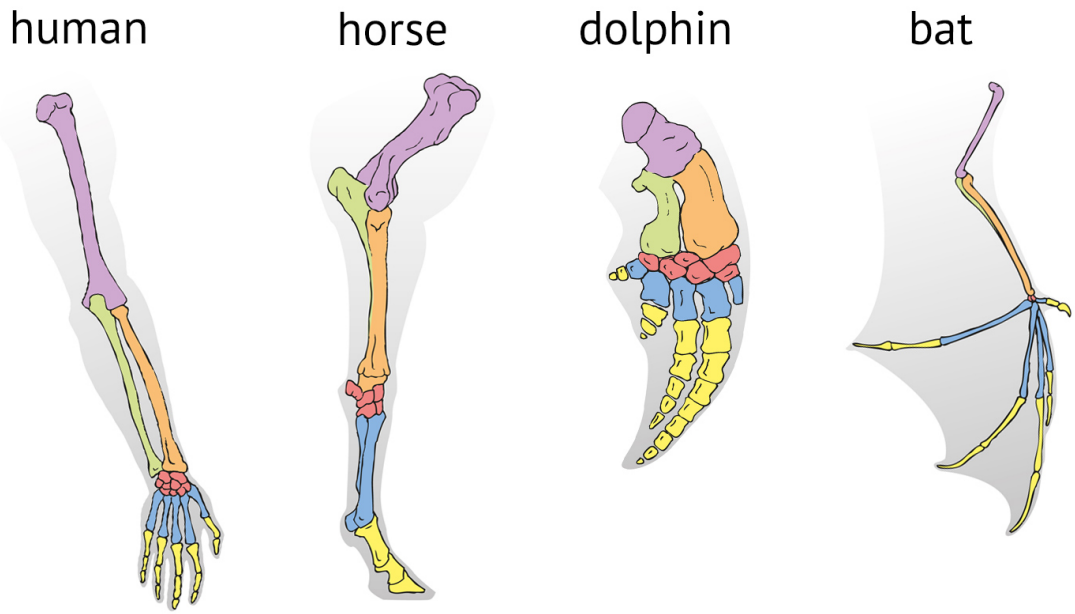
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Question 9 (3 marks)

Many living organisms have basic structures with similarities, such as the Pentadactyl limbs shown below:



Describe the term Pentadactyl limb and explain how the similarities in each organism relates to the theory of evolution. **3**

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Question 10 (4 marks)

Explain how the following evidence supports the Theory of Evolution.

(a) Fossils **2**

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(b) Geographic variation **2**

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End of Section I

Section II – Chemistry

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions 5 marks

1. An element on the periodic table is represented using the following notation:

$$^{X}_{Y}Z$$

Which of the following is true?

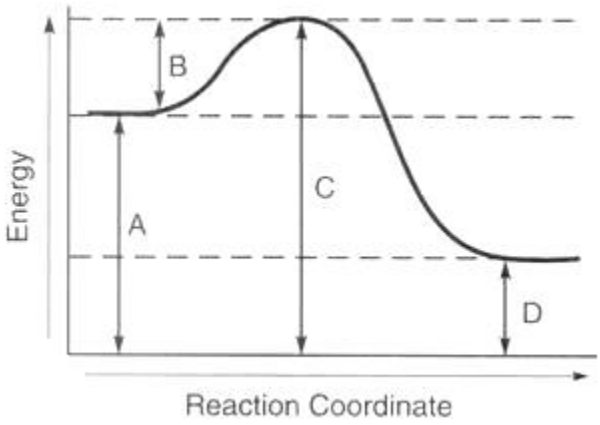
A X is the atomic number
B X is the mass number
C X is the number of protons
D X is the number of neutrons

2. A pure sample of an element is shiny, soft and reacts with oxygen at room temperature.

What group is this element most likely to belong to?

A Alkali metals
B Transition metals
C Halogens
D Noble gases

Use the information provided in the Energy Profile diagram to answer questions 3 and 4.



3. Which of the following statements is true?

A The reaction is exothermic and the reaction mixture would heat up.
B The reaction is endothermic and the reaction mixture would heat up.
C The reaction is exothermic and the reaction mixture would cool down.
D The reaction is endothermic and the reaction mixture would cool down.

4. The rate of reaction is determined by which of the following?

A A
B B
C C
D D

5. Which of the following is an example of a neutralisation reaction?

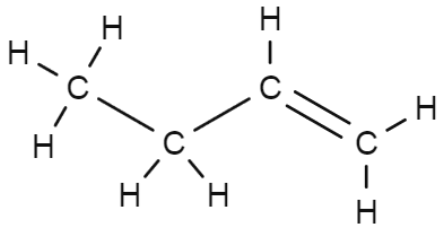
A Hydrogen ions and electrons forming hydrogen atoms.
B Hydrogen chloride reacting with ethylene to produce chloroethane.
C Hydrochloric acid reacting with lead nitrate to form solid lead chloride.
D Hydrochloric acid reacting with magnesium hydroxide to produce magnesium chloride and water.

Part B – Extended Response Questions 15 marks

Question 6 (3 marks)

Nomenclature allows chemists to communicate the structure and names of molecules.

(a) Name the following molecule: **1**



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(b) Draw the structure for a molecule of 2-bromo-3-methylheptane in the space provided. **2**

Question 7 (3 marks)

Write a balanced chemical equation, with states, for the reaction of nitric acid with calcium carbonate chips to produce calcium nitrate, carbon dioxide, and water. **3**

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Question 8 (2 marks)

Outline the similarities and differences between the bonding present in MgO and CO. **2**

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Question 9 (2 marks)

For ONE named polymer, describe a property that is related to its use. **2**

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Question 10 (2 marks)

A student places a piece of zinc metal into a solution of copper sulfate and makes observations.

(a) Describe ONE observation. **1**

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(b) Identify whether the observation is the consequence of an oxidation half reaction or reduction half reaction. **1**

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Question 11 (3 marks)

Outline the contribution of ONE named scientist to the development of atomic theory. **3**
Include a labelled diagram of their atomic model in your response.

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End of Section II

Section III – Physics

Name: .....

Class: .....

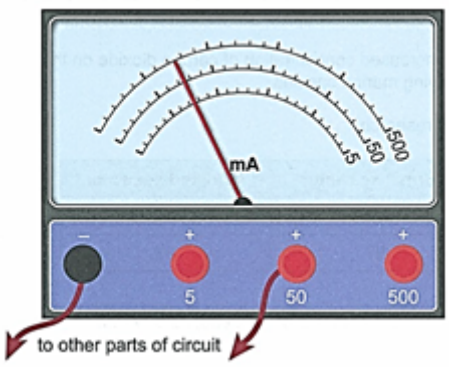
Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions **5 marks**

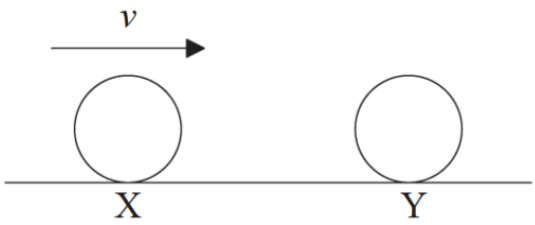
1. The diagram provided shows an electrical ammeter with multiple scales. This ammeter has been connected to a circuit with current flowing.



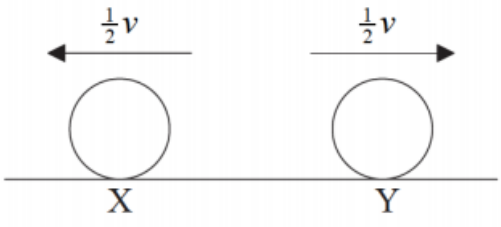
What is the reading (in mA) shown on the ammeter?

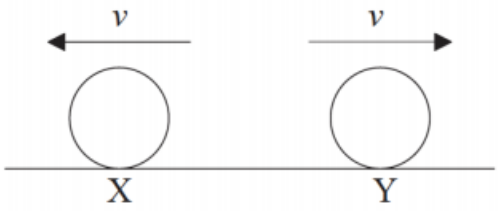
A 1.5
B 12.5
C 15
D 125

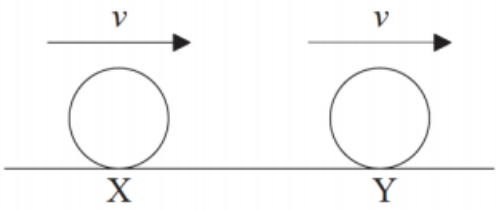
2. A simulation is setup where ball X is moving on a frictionless surface in a horizontal line to collide with an identical ball Y that is at rest.

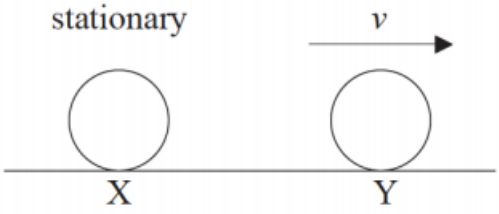


When ball X strikes ball Y with speed v, what is a possible outcome of the collision?

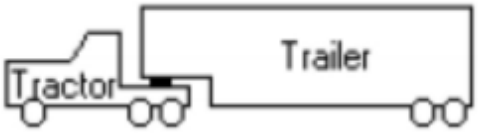
A 

B 

C 

D 

3. A tractor-trailer truck is travelling down a straight road. The tractor has a mass of m, which is connected to a trailer with a mass of 4m.



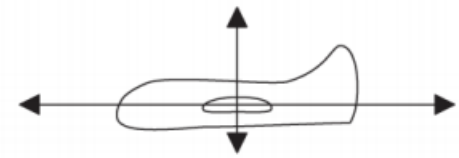
m **4m**

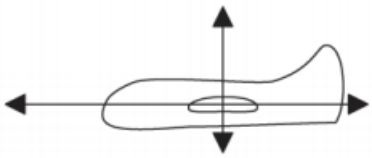
What is the force applied by the trailer to the tractor, when the tractor accelerates forward?

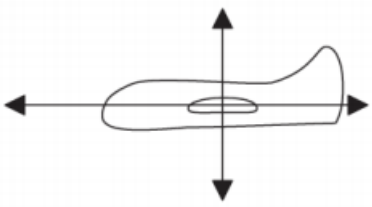
A ¼ the force of the tractor on the trailer.
B equal to the force of the tractor on the trailer.
C 4 times the force of the tractor on the trailer.
D 5 times the force of the tractor on the trailer.

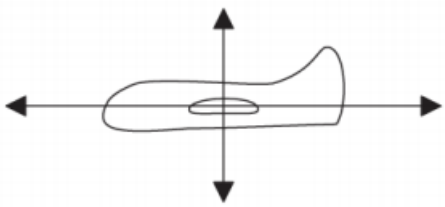
4. A model plane flies with constant velocity at a constant height.

Which diagram represents the forces acting upon the plane?

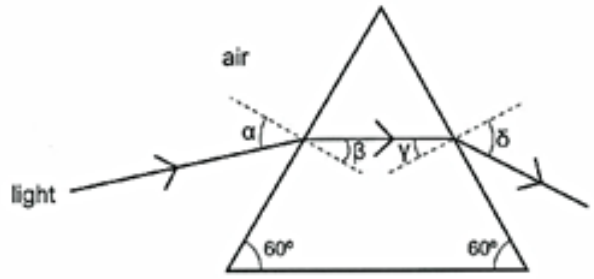
A 

B 

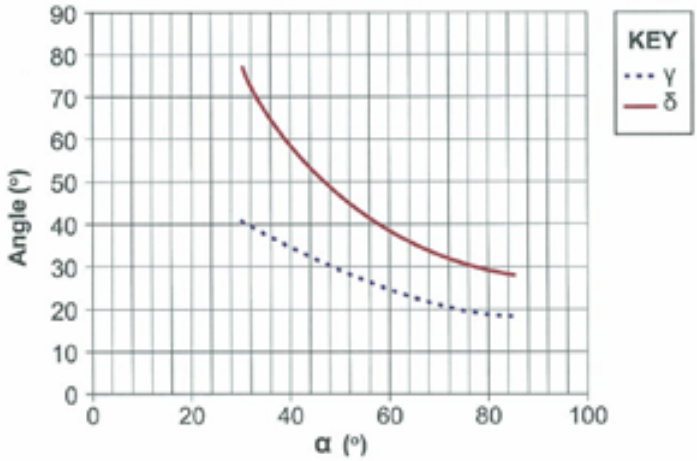
C 

D 

5. Anne drew a diagram to show the path a ray of light would take through a prism.



She varied the angle of incidence (α) whilst measuring the other angles each time and graphed her results.



As she moved the light source in an arc to vary α, she made sure that the light ray struck the prism at the same point on the surface.

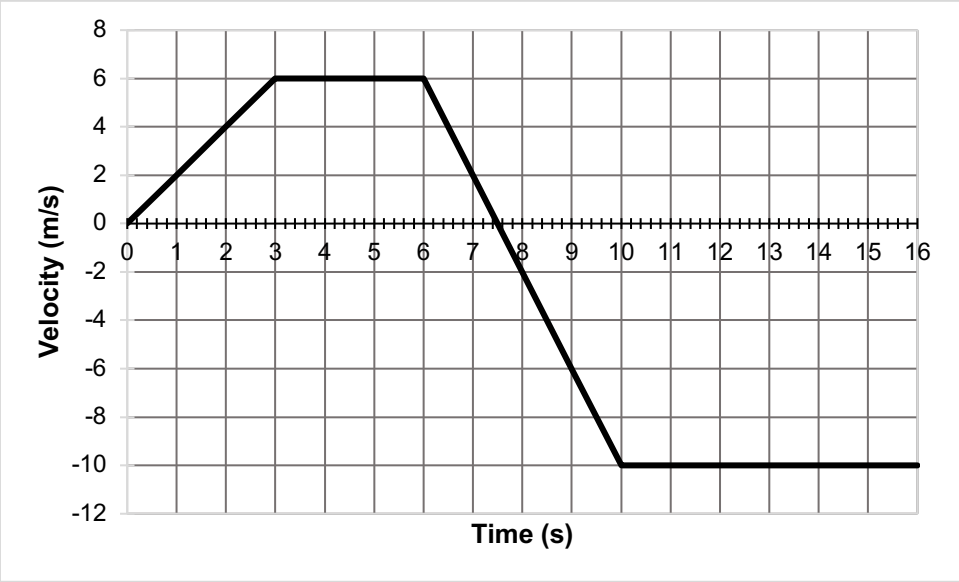
Which of these is an observation that Anne could have made during her investigation?

A The angle α is always larger than the angle γ.
B An increase in angle β produces an increase in angle γ.
C The light rays always leave the prism at the same point.
D The angle δ can reach a value of 90 degrees but the angle γ cannot.

Part B – Extended Response Questions 15 marks

Question 6 (4 marks)

Shown below is a velocity-time graph for a motorcyclist's journey.



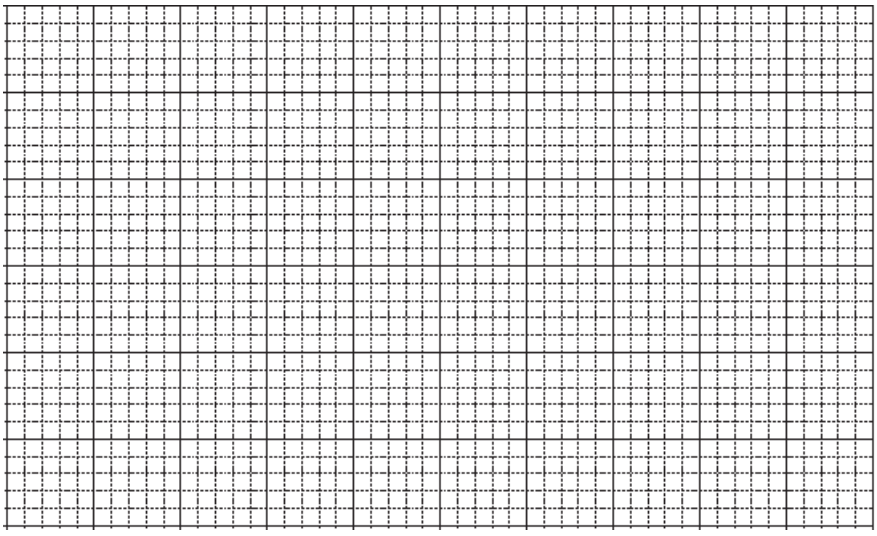
(a) Calculate the total distance travelled by the motorcyclist. **2**

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(b) Draw an acceleration-time graph of the motorcyclist's journey on the grid provided. **2**



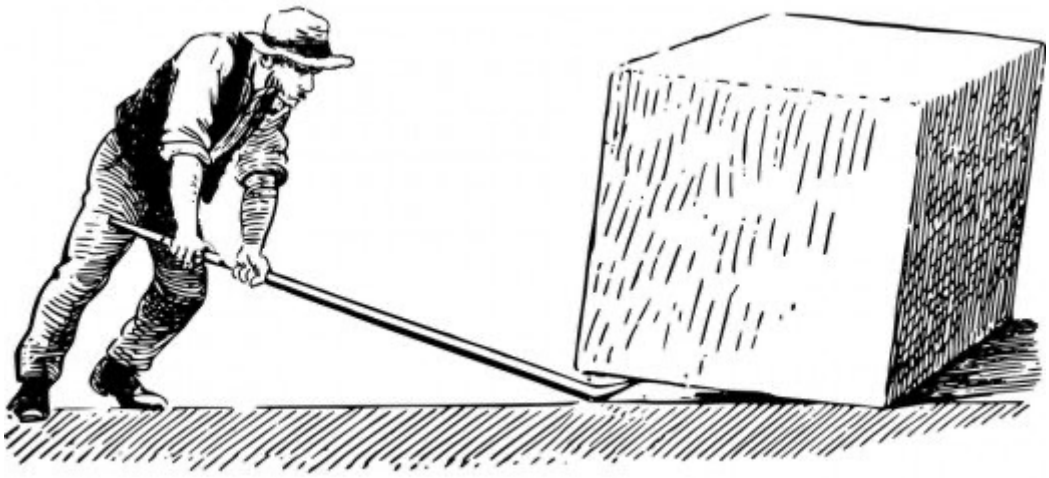
Question 7 (4 marks)

Complete the table below **4**

	Longitudinal waves	*Transverse waves*
Description	…… …… …… …… …… ……	…… …… …… …… …… ……
Example	…… ……	…… ……

Question 8 (3 marks)

Identify the simple machine present in the diagram below and explain how it works. **3**



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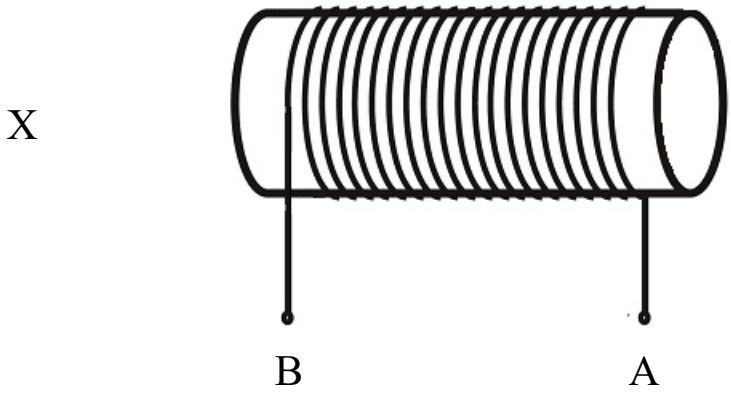
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Question 9 (3 marks)

A group of students constructed a solenoid by winding electrical wire around a hollow cardboard tube as shown in the diagram below. **4**



An electrical current was passed through this solenoid from point A to point B.

(a) Draw the magnetic field lines around the solenoid in the diagram above. **2**

(b) Indicate the polarity of the magnetic pole generated at point X. **1**

End of Section III

Section IV – Earth and Space

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions 5 marks

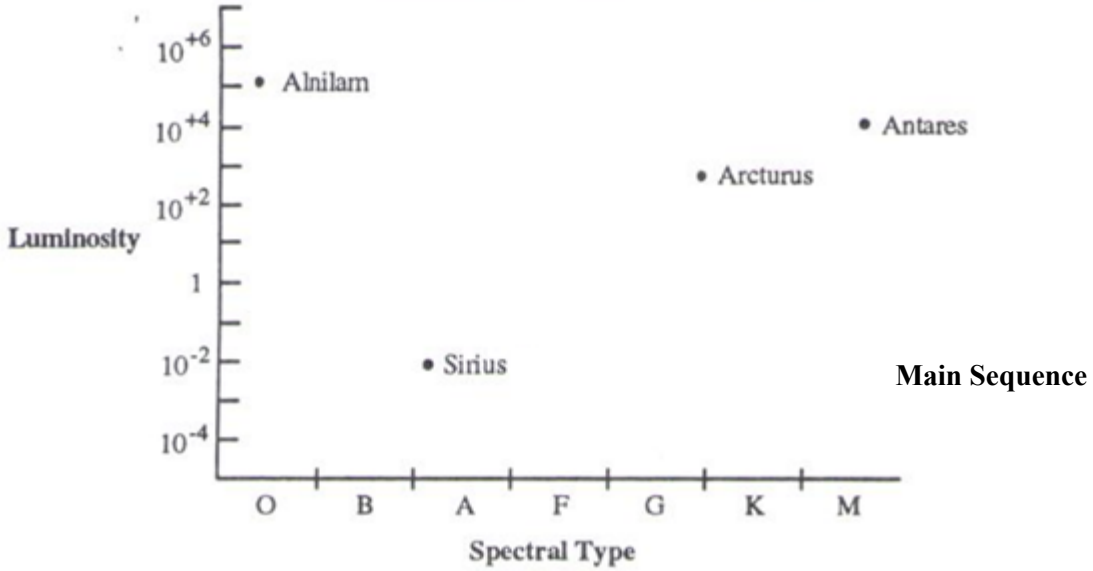
1. The crust and upper mantle are components of which of the following?

A Lithosphere
B Atmosphere
C Hydrosphere
D Asthenosphere

2. According to Wegener's model for plate tectonics, what evidence did glaciers leave for the existence of Pangaea?

A Striation patterns
B Sea-floor spreading.
C Glossopteris fossils
D Mid-ocean ridges

Use the information provided in the Hertzsprung Russell diagram to answer questions 3 and 4.



3. Which star in the diagram above has the greatest surface temperature?

A Sirius
B Antares
C Alnilam
D Arcturus

4. Which star in the diagram above is most similar to the Sun?

A Sirius
B Antares
C Alnilam
D Arcturus

5. The planet Mars is located approximately 228 million km from the sun.

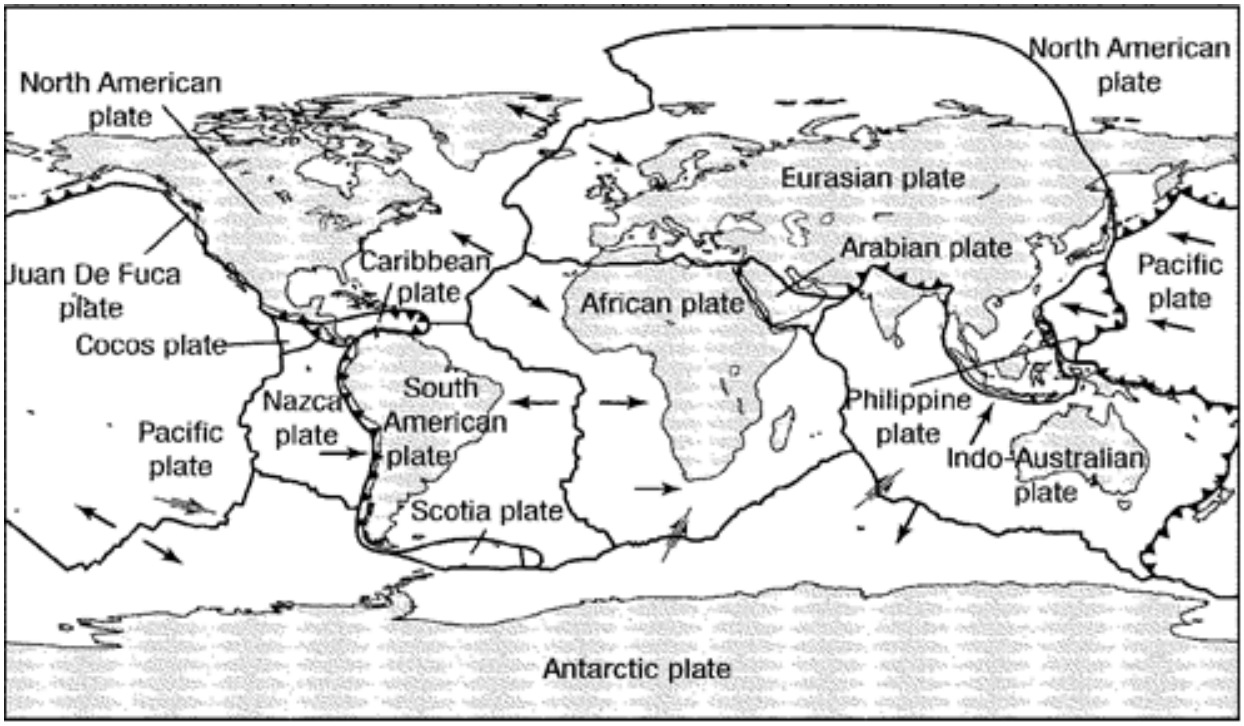
If the Sun were to release an intense burst of light in the form of a solar flare, how long would it take for this event to be observed on Mars given the speed of light is 3.0×10^8 ms^{-1}.

A 1.3 seconds
B 7.6 seconds
C 130 seconds
D 760 seconds

Part B – Extended Response Questions 15 marks

Question 6 (3 marks)

The diagram below illustrates the plate boundaries for the different continental plates.



(a) Identify the type of plate boundary between the South American and Nazca plates. **1**

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(b) Describe the features that would be observed at this plate boundary identified in (a) **2**

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Question 7 (2 marks)

Calculate how much energy is released in a nuclear reaction where 4.37×10^{-28} g of mass is lost using Einstein's mass-energy equivalence formula ($\mathbf{E = mc^2}$). **2**

Show all necessary working.

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Question 8 (4 marks)

The stellar evolution of a star is dependent upon its mass during the main sequence phase of its lifespan. Predict the stellar evolution pathway for the following:

(a) A star with a similar mass to the sun. **2**

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(b) A star weighing up to 20 solar masses. **2**

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Question 9 (2 marks)

Compare the density of rock samples obtained from the oceanic and continental crust. **2**

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Question 10 (4 marks)

The Big Bang theory is the leading explanation for the origins of our universe.

Explain how the following pieces of evidence support the expanding universe model.

(a) Cosmological Red Shift **2**

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(b) Cosmic Microwave Background Radiation. **2**

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End of Examination